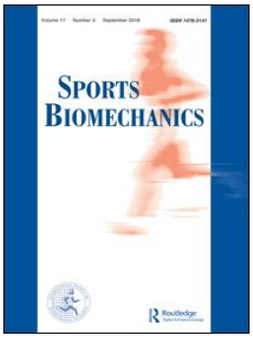

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Influence of forefoot bending stiffness on American football performance and metatarsophalangeal joint bending angle

John William Wannop, Nicole Schrier, Jay Worobets and Darren Stefanyshyn

Human Performance Lab, Faculty of Kinesiology, University of Calgary, Calgary, Alberta, Canada

ABSTRACT

Changes in forefoot bending stiffness have been shown to affect metatarsophalangeal peak bending angles as well as athletic performance. Increasing bending stiffness tends to reduce peak bending angles, which could potentially reduce hyperextension injuries such as turf toe. Limited information is available, however, on the efficacy of increasing forefoot bending stiffness on large-sized athletes such as those that participate in American Football, with prior studies being conducted on smaller athlete populations. Therefore, the purpose of this study was to determine the influence of increased forefoot bending stiffness on metatarsophalangeal joint extension and athletic performance of grid-iron football players. Ten varsity grid-iron football players performed four National Football League combine movements in a motion capture laboratory in three footwear conditions of varying bending stiffness: Soft (12.7 N/mm), Control (23.8 N/mm), Stiff (42.2 N/mm). None of the footwear conditions significantly altered the maximum metatarsophalangeal bending. Therefore, to reduce metatarsophalangeal hyperextension injuries in American football players a greater amount of forefoot bending stiffness may be required. Performance differences were present only during the five-metre sprint acceleration, with athletes having an improved performance in the Control and Stiff conditions. This improved performance was due to an increased horizontal impulse and improvements in power generation at the ankle joint.

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Introduction

The metatarsophalangeal (MTP) joint is in the forefoot region of the foot, being comprised of five metatarsals and five proximal phalanges, creating five individual condyloid joints. The rounded surface of each metatarsal bone articulates with the elliptical cavity of the proximal phalange, providing movement in two planes (primarily flexion/extension, with a small amount of abduction/adduction). Although the MTP joint consists of five individual joints, within the biomechanics community, it is generally simplified as a single hinge joint (Stefanyshyn & Wannop, 2016).

Many studies have investigated the influence that manipulating the MTP joint can have on performance with increases in forefoot bending stiffness leading to improvements in running economy (Roy & Stefanyshyn, 2006), sprinting (Stefanyshyn & Fusco, 2004), jumping (Stefanyshyn & Nigg, 2000; Tinoco et al., 2010) and cutting/lateral movements

CONTACT John William Wannop b.wannop@ucalgary.ca
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(Tinoco et al., 2010; Vienneau et al., 2016; Worobets & Wannop, 2015). The exact mechanism is not fully understood but has been speculated to be caused either by shifting the point of application of the ground reaction force (Willwacher et al., 2013) potentially leading to an optimisation of the athletes' musculoskeletal properties (Madden et al., 2016) and/or aspects involving a reduction in the amount of energy lost at the MTP joint during a specific movement (Stefanyshyn & Nigg, 2000; Tinoco et al. 2010).

Aside from potential performance benefits, the metatarsophalangeal joint (MTP) has been an important athletic injury research topic over the past decade (Crandall et al., 2015; Ford et al., 2015; Frimenko et al., 2012, 2013; Riley et al., 2013). Due to the large number of bones, muscles and ligaments many injuries may occur within the forefoot.

American football is popular worldwide, consisting of a multitude of athletic movements such as pivoting, sprinting, running, jumping and cutting. These movements performed at maximum intensity place large loads on the athletes and since American football also involves direct contact between players, the likelihood of overload and risk of injury is high. In American football sprain of the first MTP joint can occur and is often referred to as turf toe. One of the most common mechanisms of MTP joint sprain, accounting for over 85% of turf-toe injuries in American football is hyperextension of the MTP joint (Coker et al., 1977). This hyperextension results in damage to the plantar structures of the forefoot, specifically injury to the flexor halluces brevis, plantar capsule ligament and in extreme cases, injury to the sesamoid bones within the tendon. While the injury rate of MTP sprain is not substantial, being reported at 4.5–6.0 injuries per football season per team (Clanton et al., 1986; Coker et al., 1977), and accounting for less than 2% of all injuries (Turbeville et al., 2003) it can be severe, resulting in a large time loss for the athlete. It has been reported that although ankle injuries are four times more prevalent than turf toe injuries, turf toe injuries account for over 4.5 more games lost per injury (Coker et al., 1977).

While research has shown that increasing the forefoot bending stiffness of athletes may lead to an improved performance and may reduce the range of motion of the MTP joint, thereby reducing the potential for forefoot hyperextension injury to occur, these previous studies were conducted on smaller athletes performing on stiff surfaces, such as a gymnasium or running track. No information is available of the efficacy of increases in forefoot bending stiffness on large athletes, such as those that participate in American football while performing on compliant turf surfaces. Towards this end, it is unclear as to what magnitudes of forefoot stiffness may be required to reduce the MTP bending of these larger athletes or how effective common increases in MTP stiffness are at improving performance and reducing excessive joint extension during common football movements. Therefore, the purpose of this study was to determine the influence of increased forefoot bending stiffness on MTP extension and athletic performance of grid-iron football players. It is hypothesised that increasing the forefoot bending stiffness would reduce MTP extension and increase athletic performance.

Methods

Footwear

To investigate the effect of forefoot bending stiffness on performance and MTP joint kinematics, three custom footwear conditions were created consisting of prototype

adidas football cleats (size = US12) with different amounts of forefoot bending stiffness and all other aspects being kept identical (Figure 1). The forefoot bending stiffness of the three footwear conditions were measured using a three-point bend test performed by the manufacturer. The three-point bending test was performed on the entire shoe (Ford et al., 2016; Roy & Stefanyshyn, 2006). The forefoot of the shoe was placed on a frame, which had two supporting points (80 mm apart). A stamp 12 mm thick and 70 mm wide was attached to a material testing machine that measures force and displacement (Instron Corp., Canton MA., USA, model #8502). The machine was set to displace the centre point of the forefoot 7.5 mm vertically downward in a time of 0.1 s, at which point it returned to its original position in the same amount of time (0.1 s). This cycle was repeated 20 times and the mean force required to displace the stamp from 5 to 6 mm was used as the longitudinal bending stiffness of the shoe (N/mm). The stiffnesses of the shoes were Soft (12.7 N/mm), Control (23.8 N/mm) and Stiff (42.2 N/mm) conditions. This range of stiffness was selected to span from the lower end to the higher end of stiffness values of previously published studies (Ford et al., 2016; Nguygen et al., 2015; Roy & Stefanyshyn, 2006; Stefanyshyn & Wannop, 2016).

Figure 1. Photograph of the footwear model used during testing. All footwear conditions were identical except for forefoot bending stiffness.

Motion analysis

Data were collected on ten university grid-iron football players (mean ± SD, height = 1.86 ± 0.05 m, mass = 96.2 ± 14.2 kg). Before participating in the study all participants were required to read and sign a participant consent form approved by the University of Calgary ethics committee. To be included in the study all participants had to be a member of the university football team, free from recent lower extremity injury and properly fit the footwear.

An eight-high speed digital video camera system (Motion Analysis Corp., Santa Rosa, CA, USA) sampling at a frequency of 240 Hz was used to record the three-dimensional trajectories of retro-reflective markers placed on the lower leg of the athlete, while a force platform embedded in the ground (Kistler AG, Winterthur, Switzerland) concurrently recorded three-dimensional force data at a frequency of 2400 Hz. Three-dimensional kinematics of the lower limb were recorded for the right leg of each athlete during testing. The forefoot, rearfoot, and shank were defined by attaching retro-reflective markers, measuring 19 mm in diameter to each segment using double sided tape. Three markers per segment were used with markers being attached in the following locations: fibula, upper tibial crest, distal lateral leg to define the shank segment, posterior shoe heel, distal shoe heel, lateral shoe heel to define the rearfoot segment, and medial toe, middle toe and lateral toe to define the forefoot segment. The motion capture system was calibrated to an

accuracy defined by a 3D residual below 0.6 mm. To determine the locations of the metatarsophalangeal, ankle and knee joint centres, additional markers were placed on the athlete and a standing neutral trial was recorded. The athlete was asked to stand with their feet hip width apart and the knee and hip fully extended. To determine the MTP joint centre, additional markers were placed on the base of the 1st and base of the 5th metatarsal. For the ankle joint centre additional markers were placed on the medial and lateral malleoli, while for the knee joint centre additional markers were placed on the lateral aspect of the knee and on the centre of the patella. Prior to analysis, the kinematic and kinetic data was filtered using a low-pass fourth order Butterworth filter with cut-off frequencies of 12 Hz and 100 Hz respectively. The kinematic and kinetic data were imported into Kintrak 7.0.25 (University of Calgary, Calgary, Canada) for analysis with the resultant joint moments and angular impulses in all three planes calculated using an inverse dynamics approach.

Artificial turf installation

The laboratory was set up to allow kinetic and kinematic data collection on an artificial turf surface. The artificial turf installed in the laboratory was a third-generation infilled surface (Ecofill Pro Series 3NX FTS from Mondo Worldwide) and was installed according to the manufacturers' specifications. The turf was installed in all areas of the lab, including the length of the lab runway. The Ecofill Pro Series turf consists of long monofilament fibres, each with a semi concave shape and a height of 45 mm as well as a sand/rubber infill combination. To increase compliance, the turf was placed over a 23 mm underlying shockpad composed of post-consumer rubber granules completely covered with a non-woven material. To ensure accurate kinetic data collection, a separate piece of shockpad and infilled turf were cut to fit inside a 60 cm x 90 cm metal tray that was rigidly bolted to the force platform. After installation, mechanical properties of the artificial turf were quantified using a Clegg Impact Hammer and Penetrometer to ensure the installed turf met the manufacturers' specifications.

Movements

During testing, each athlete completed four different movements in each of the footwear conditions, with the movements being derived from common National Football League (NFL) combine drills. The four drills consisted of: 5-10-5 agility drill (Agility), 5 m sprint acceleration from a three-point stance (Sprint), maximum effort vertical jump (V-Jump), and a maximum effort horizontal jump (H-Jump). For the Agility drill, athletes performed three sprint and cut manoeuvres initiating from a three-point stance: a 5-yard sprint (4.57 m), 180° turn (which took place on a force plate), 10-yard sprint (9.14 m), 180° turn and a 5-yard (4.57 m) sprint (Figure 2). Performance was defined as the total time it took the athletes to complete the drill.

The maximum effort horizontal jump consisted of a counter movement jump with the goal of the athletes to travel as far horizontally as possible. Athletes stood upright with their right foot in the centre of a force platform, then performed a single controlled countermovement and jumped horizontally forward as far as possible (Figure 3, left). For a trial to be accepted, the athletes were required to land on their feet and were not allowed

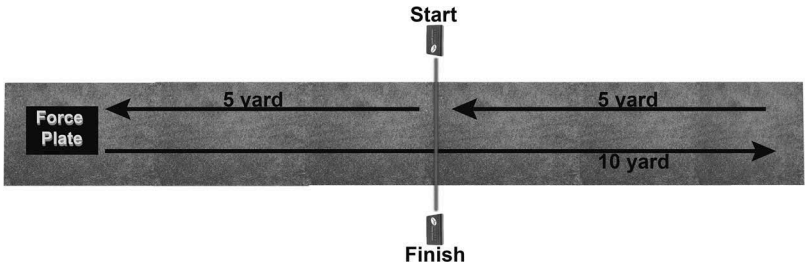

Figure 2. Schematic of the 5-10-5 agility drill.

to move (they were required to maintain balance or footing) or fall during landing. Performance was determined as the change in horizontal distance during the jump of the distal shoe heel marker. The maximum effort vertical jump was similar to the horizontal jump except the goal was for the athletes to reach as high vertically as possible. Athletes again stood upright with their right foot placed in the centre of a force platform and then performed a countermovement jump reaching as vertically high as possible and striking a Vertec jump height measurement system (Sports Imports, Hilliard, OH, USA). The performance variable was defined as the maximum jump and reach height that the athlete was able to attain (Figure 3, centre). Lastly, the athlete performed a 5-metre sprint acceleration from a three-point stance (Figure 3, right). The athlete started in a three-point stance, with the starting line placed directly in front of their downed hand and they were positioned to ensure that their second footfall was in the centre of the force platform. The performance outcome was defined as the time it took the athlete to travel 5 metres measured with photocell timing lights (Brower Timing Systems, Draper, UT, USA). Athletes completed five successful trials for each movement in each footwear condition with both the order of footwear conditions worn within each drill and the order of the individual drills being randomised between participants.

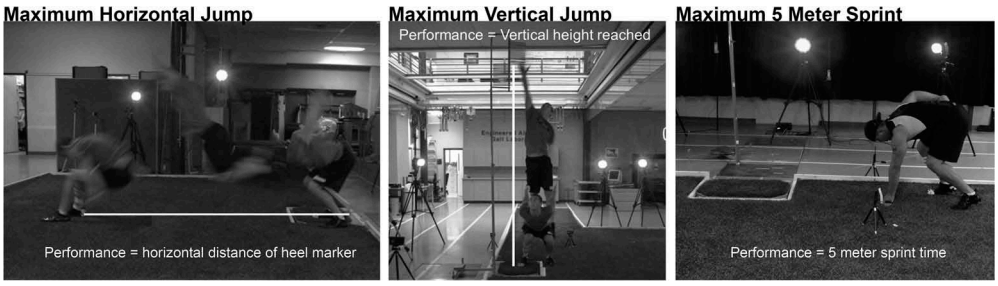

Figure 3. Photographs of the maximum horizontal jump (left), maximum vertical jump (centre) and the maximum 5-metre sprint acceleration (right).

Statistics

The primary variables of interest were the maximum amount of MTP extension and the performance measures for each movement (5-m sprint time, agility drill time, vertical jump and reach height and long jump distance). A one-way analysis of variance

(ANOVA) was used to determine any differences in performance or maximum MTP extension due to footwear condition for each of the four drills, with the level of significance set to α = 0.10. If differences were found within the primary variables of interest, secondary variables of interest were investigated including the ground reaction forces, ankle joint moments, angular impulses, joint powers, joint work and knee joint moments. Effect size (ES) calculations were made following the method of Cohen (1992) and were reported if a significant difference was present. All statistical analyses were performed using SPSS software version 20.0 (SPSS Inc., Chicago Illinois).

Results

Performance results and peak MTP extension angles for all movements can be seen in Table 1. A significant change in performance occurred during the 5-m sprint (F = 3.02, p = 0.070, ES = 0.25), with post-hoc analysis indicating an improved performance in both the Control (p = 0.085) and the Stiff (p = 0.094) conditions compared to the Soft. No significant differences were observed during any of the other movements. No statistically significant differences in the peak MTP extension angle were measured between footwear conditions during any of the movements.

Due to performance differences only occurring during the 5-m sprint, additional analysis of kinematic and kinetic variables was performed solely for this movement (Tables 2 and 3). During the 5-m sprint, there was a significant increase in the horizontal ground reaction impulse as forefoot bending stiffness was increased (F = 5.49, p = 0.014, ES = 0.12), with increases occurring in the Control shoe compared to the Soft (p = 0.013), while the Stiff condition had the trend of an increased horizontal ground reaction force compared to the Soft (p = 0.104). The ankle joint sagittal plane angular impulse was also significantly increased as stiffness was increased (F = 4.46, p = 0.027, ES = 0.33), with the Control condition resulting in an increased ankle joint angular impulse compared to the Soft (p = 0.022). No other significant differences in kinetic variables were present.

The ankle kinematics, joint work and joint power during the 5-m sprint are shown in Table 3. No difference in peak ankle dorsiflexion angle was present regardless of footwear condition, however, both the net ankle joint work (F = 5.536, p = 0.015, ES = 0.41) and the ankle joint peak power (F = 7.876, p = 0.004, ES = 0.50) were significantly altered with increased bending stiffness. Specifically, the Control condition significantly increased both the net ankle joint work and the ankle joint peak power when compared to the Soft

Table 1. Performance and peak MTP extension results during the four NFL combine drills in the three different footwear stiffness conditions. Data represent the mean of all athletes, with 90% confidence intervals in parenthesis. Significant differences (p < 0.10) are highlighted in bold with effect size in parenthesis.

	Movement	Soft	Control	Stiff	F	p-value (ES)
Performance	Agility Drill [s]	7.53 (7.29–7.76)	7.56 (7.30–7.83)	7.60 (7.35–7.86)	0.44	0.650
	5 m Sprint [s]	1.22 (1.14–1.30)	1.16 (1.11–1.21)	1.16 (1.09–1.23)	3.02	**0.070 (0.25)**
	Broad Jump [m]	2.40 (2.22–2.59)	2.41 (2.24–2.58)	2.39 (2.23–2.55)	0.11	0.890
	Vertical Jump Height [m]	3.18 (3.10–3.26)	3.17 (3.09–3.25)	3.17 (3.09–3.25)	0.35	0.710
MTP Extension	Agility Drill [deg]	11.5 (8.9–14.2)	13.2 (10.5–16.0)	12.8 (10.2–15.3)	0.88	0.430
	5 m Sprint [deg]	34.2 (31.9–36.5)	33.0 (30.7–35.3)	31.8 (28.7–35.0)	2.11	0.154
	Broad Jump [deg]	46.4 (42.3–50.5)	40.6 (38.1–43.2)	42.8 (38.7–46.9)	2.17	0.134
	Vertical Jump Height [deg]	16.8 (13.7–19.9)	16.4 (13.5–19.2)	17.1 (14.6–19.7)	0.06	0.940

Table 2. Kinetic data during the sprint movement in the different footwear stiffness conditions. Data represent the mean of all athletes with 90% confidence intervals in parenthesis. Significant differences (p < .10) are highlighted in bold with effect size in parenthesis.

Variable	Soft	Control	Stiff	F-value	p-value (ES)
Peak Horizontal Ground Reaction Force [N]	642 (602–682)	646 (601–691)	633 (584–683)	1.48	0.253
Horizontal Ground Reaction Impulse [Ns]	73.3 (68.1–78.5)	76.3 (70.6–81.9)	74.7 (69.3–80.1)	5.49	**0.014 (0.12)**
Peak Vertical Ground Reaction Force [N]	1709 (1581–1837)	1688 (1544–1833)	1685 (1530–1839)	0.64	0.537
Vertical Ground Reaction Impulse [Ns]	234 (205–263)	240 (210–271)	239 (210–269)	1.20	0.323
Peak Ankle Plantarflexion Moment [Nm]	292 (264–320)	300 (270–330)	291 (264–317)	1.10	0.353
Ankle Angular Impulse [Nms]	38.9 (35.1–42.7)	40.9 (36.7–45.1)	39.4 (35.7–43.1)	4.46	**0.027 (0.33)**
Peak Knee Extension Moment [Nm]	62 (33–90)	71 (41–102)	69 (46–93)	0.28	0.763
Knee Angular Impulse [Nms]	6.5 (2.9–10.1)	7.4 (3.4–11.4)	7.1 (4.8–9.4)	0.21	0.816
Stance Time [s]	0.214 (0.199–0.230)	0.221 (0.208–0.234)	0.219 (0.206–0.231)	1.65	0.220

Table 3. Ankle joint kinematics, joint work and joint power during the 5-m sprint in the different footwear stiffness conditions. Data represent the mean of all athletes with 90% confidence intervals in parenthesis. Significant differences (p < .10) are highlighted in bold with effect size in parenthesis.

Variable	Soft	Control	Stiff	F-value	p-value (ES)
Peak Ankle Dorsiflexion Angle [deg]	24.1 (19.5–28.7)	26.5 (22.2–30.9)	25.5 (21.3–29.6)	2.406	0.122
Ankle Net Work [J]	65.1 (55.9–74.4)	71.3 (63.5–79.1)	63.6 (54.8–72.4)	5.536	**0.015 (0.41)**
Ankle Peak Power [W]	2282 (1924–2642)	2433 (2046–2821)	2181 (1814–2548)	7.846	**0.004 (0.50)**

(p = 0.038 for work and p = 0.054 for power) and when compare to the Stiff condition (p = 0.003 for work and p = 0.003 for peak power).

Discussion and implications

The purpose of this study was to determine the influence of increased forefoot bending stiffness on peak MTP bending angle and athletic performance of grid-iron football players. While previous studies have shown that athletes may have improved performance and reduced MTP bending with increased forefoot bending stiffness, these previous studies were conducted on smaller athletes performing on stiff surfaces, such as a gymnasium or running track. The results of this study provide insight into the magnitudes of forefoot stiffness that are required to reduce the MTP bending of large grid-iron football players performing on compliant surfaces and how current magnitudes of forefoot bending stiffness incorporated into footwear influence performance of these large athletes.

Overall, when forefoot bending stiffness was increased, no significant changes in MTP bending angle occurred for any movement and no performance improvements were present, with the exception of during the 5-m sprint acceleration. It should be specifically

noted that during large ranges of motion (> 30°, during the 5-m Sprint, Broad Jump) subtle trends ($p = 0.154$ and $p = 0.134$) were present as on average, peak MTP extension angles were reduced with increased bending stiffness, although this result was not statistically significant. Therefore, to reduce the risk of hyperextension injury to the MTP joint through a reduced peak MTP bending angle in large athletes (grid-iron football players) performing sprinting, cutting and jumping movements, forefoot bending stiffness of greater than 42.2 N/mm may be required. A previous study showed a reduction in the peak MTP bending angle using footwear with stiffness values lower than this amount (Smith et al., 2016), however, this study measured the MTP bending angle later into the sprinting movement. While space limitations prevented similar measurements in the current study, it may be the case that during the early steps of a sprint acceleration where the amount of MTP bending is reduced, the influence of the increased forefoot bending stiffness is not as substantial. Additionally, the participants in the current study were substantially larger than previous studies (mean mass = 96.2 kg, shoe size 12 US vs mean mass = 73.8 kg, shoe size = 9 US) and potentially, the strength of the larger football players were sufficient to maintain the normal range of motion in the footwear conditions with increased forefoot bending stiffness. It could, therefore, be proposed that the athlete population and anthropometrics should be considered when determining the optimal magnitude of forefoot bending stiffness in footwear, additional considerations should also be made regarding the surface.

While footwear above 42.2 N/mm may reduce potential MTP hyperextension injury risk in grid-iron football players, stiffness values equal to and lower than this did result in an increased performance during the 5-m sprint acceleration. Both the Control and Stiff shoe conditions improved sprint acceleration performance, which supports previous research where sprinters had an increased 20 m sprinting performance in stiff footwear (Stefanyshyn & Fusco, 2004). While both the Control and Stiff shoe conditions improved sprinting performance, this result may have been achieved through different mechanisms for each condition. The Control footwear condition increased the horizontal ground reaction impulse (4.1%) while increasing the force and power generating capability of the ankle joint, increasing the angular impulse (5.1%), peak power (6.6%) and net joint work (9.5%) compared to the Soft. Therefore, the stronger football athletes had an ability to increase the output from their ankle joint plantarflexors to overcome the increased bending stiffness of the Control shoe condition, increasing the ankle joint peak power and net work while maintaining a similar amount of MTP extension.

When sprinting in the Stiff shoe condition, no group differences at the ankle joint were observed, however, there was the trend of an increase in the horizontal ground reaction impulse in this condition. From the kinematic and kinetic data it is not clear why athletes had improved sprint acceleration performance in the Stiff shoe condition. It may be that the performance variables were altered during other footfalls of the 5–m sprint which were not measured. It has been proposed that different stiffness of footwear would be needed for different phases of a sprint acceleration (Toon, 2008), therefore, perhaps the Control condition was ideal during the initial steps (which were measured by the force plate and motion capture system) while the Stiff condition was ideal when exiting the three-point stance or during later steps (which were not measured). Future studies will aim to investigate these possibilities and gain insight into the mechanism of these performance increases. Furthermore, it has also been

proposed that athletes may employ different strategies to exploit increased bending stiffness, with some increasing their ankle joint moments, while maintaining constant stance time, while other athletes decrease their ankle joint moments while increasing stance time (Willwacher et al., 2014). It is unclear if athletes within this study employed different strategies in the presence of increased bending stiffness and expansion of the sample size of athletes collected may assist in answering this question.

Future work

Based on the results of this study, the upper values of forefoot bending stiffness that were used (42.2 N/mm) and which has been used in previous studies (Ford et al., 2016; Nguygen et al., 2015; Roy & Stefanyshyn, 2006; Stefanyshyn & Wannop, 2016) is not sufficient to substantially reduce MTP bending of large, football athletes. It is speculated that further increases in forefoot bending stiffness are required in order to restrict MTP bending and potentially reducing MTP joint injury in grid-iron football players. However, as the forefoot bending stiffness of footwear is increased, it tends to be perceived as uncomfortable by athletes (Luo et al., 2009; Miller et al., 2000). Implementation of a non-linear stiffness element in footwear may be a method of improving athlete comfort, while providing high levels of stiffness when need. The non-linear stiffness element incorporated into the footwear may provide low levels of stiffness during low bending angles and large amounts of stiffness at large bending angles. Non-linear bending stiffness has been investigated previously, with positive results being reported in terms of kinematic and kinetic variables of soccer players (Wannop et al., 2017). It is currently unclear what the stiffness profile of this non-linear element should be, however, future studies will focus on determining this profile for grid-iron football players.

In terms of the results contained in this study, it is currently unclear how the data should be extrapolated either within the same athlete population or to other athlete populations. Specifically, it is unknown how the scaling of forefoot bending stiffness should be performed for different athlete populations and/or for different shoes sizes within the same athlete population. There is no data available on whether the magnitude of bending stiffness should be similar for different sizes of footwear (size 10US vs size 16US, for example) or whether it should be scaled and if it should be scaled, what method of scaling should be utilised (linear vs. non-linear scaling, for example). Additional information is also needed regarding how the anthropometrics of the individual athlete can be used to determine the optimal bending stiffness, with recent studies indicating that the strength profile of the ankle plantarflexors may provide insight (Madden et al., 2016; Willwacher et al., 2016). These aspects are all currently the aim of future investigations.

Conclusion

Increases in forefoot bending stiffness within currently used ranges in footwear worn by grid-iron football players did not reduce the maximum amount of MTP bending and, therefore, may be ineffective at reducing the risk of MTP hyperextension injury.

Further increases in stiffness values should be investigated to determine their efficacy at reducing MTP hyperextension injury. In terms of performance, increased bending stiffness improved performance only during the sprint acceleration movement, with athletes increasing the output of their ankle joint plantarflexors in the stiff footwear.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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